

Chapter 3: Interdependence and the Gains from Trade

Main Example

Setting

There are **two countries** (Taiwan and Cambodia), each only produce **two goods** (chip batches and tons of rice), using only one input (labor). We also assume each country switches between two goods at a constant rate, and works for **8 hours a day**. The labor conditions are given below.

	Mins per chip batch	Mins per ton of rice
Taiwan	20	10
Cambodia	60	15

We see, if each country spends 4 hours on each good each day, Taiwan can produce **12 batches of chips** and **24 tons of rice** a day, whereas Cambodia can produce **4 batches of chips** and **16 batches of rice** a day. Although any other point on the PPF is possible. Clearly, **Taiwan has an absolute advantage**.

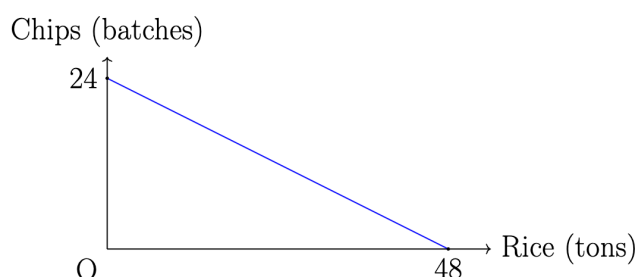


Figure 3: Taiwan's PPF

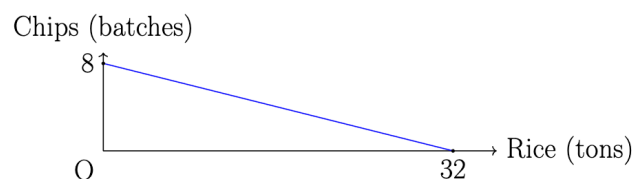


Figure 4: Cambodia's PPF

However, in terms of **opportunity cost**, we see

	Opportunity cost of 1 chip batch	Opportunity cost of 1 ton of rice
Taiwan	2 tons of rice	0.5 chip batches
Cambodia	4 tons of rice	0.25 chip batches

Then, Taiwan has a comparative advantage in chips ($2 < 4$) and Cambodia has a comparative advantage in rice ($0.25 < 0.5$).

Suppose Taiwan shifts more time towards chips, producing **18 chip batches and 12 tons of rice** a day, and Cambodia specializes completely in rice, producing **32 tons of rice and no chips**.

Suppose Cambodia trades 15 tons of rice to Taiwan in exchange for 5 chip batches (3 tons of rice per chip batch). This lies in **between Taiwan's opportunity cost of a chip batch (2 tons of rice) and Cambodia's (4 tons of rice)**, which is **exactly the range** which both countries are willing to trade.

Notice, after trading, both countries get to be above their PPFs, which is a **gain for both sides**.

Summary

Trade allows for mutual benefit due to **comparative advantages** comparing **opportunity costs**, instead of absolute advantages. Trade lets each country consume **outside its own PPF**, a combination of goods it could have never produced by itself.

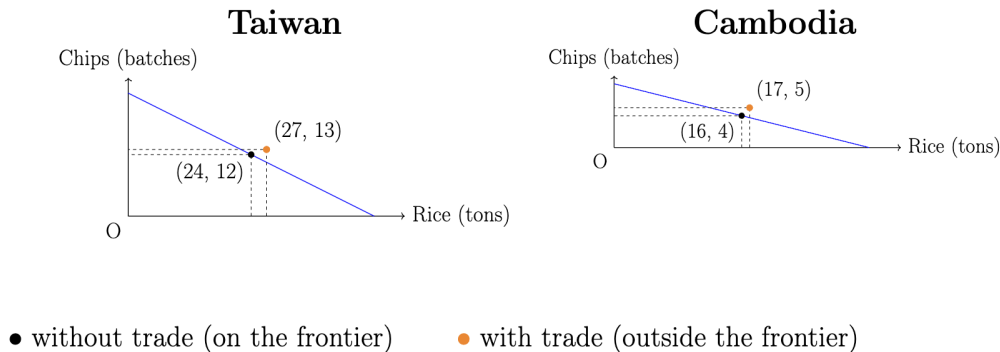


Figure 5: Taiwan and Cambodia's PPF vs after trading

Remark

It is impossible for one country to have a comparative advantage in both goods, since the opportunity cost of one good is the inverse of the opportunity cost of another. A country that is relatively cheap at producing one good is automatically relatively expensive at producing the other.

Other Examples

- The surgeon and her assistant: A surgeon who can type faster than anyone else in her office should still hire an assistant to handle her scheduling and paperwork, since every hour she spends on typing is an hour she is not spending on surgery (i.e. opportunity cost of an hour of typing is far higher)
- Taiwan's Rice Import Quota: It raises the country's total economic pie to import rice, but it **can hurt specific producers** (i.e. domestic rice farmers) $\Rightarrow$ **Sometimes it's important to protect local producers, rather than maximizing purely for consumer benefit**

Shapes of PPFs

PPFs are **straight lines** if a country *switches* between producing its two goods at **a constant rate**.

Real economies oftentimes have PPFs that **bow outward** instead of running straight, where the **opportunity cost rises** the more a good is produced.

Constant opportunity cost Increasing opportunity cost

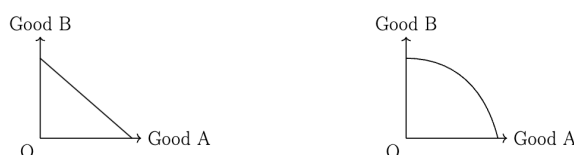


Figure 6: Comparison of different PPF shapes